

BUSINESS INTELLIGENCE AND TOOLS

EXPERIMENT - 1

Problem File

Title: Data Acquisition and Data Profiling using BI Tools

Problem Statement

Organizations collect large amounts of data from different sources such as CSV files, Excel spreadsheets, and databases. Before this data can be used for analysis and decision-making, it needs to be acquired, examined, and checked for quality issues.

The problem is to acquire data from multiple sources and perform data profiling using a Business Intelligence (BI) tool. The acquired datasets should be examined to identify missing values, duplicate records, incorrect data types, inconsistent values, and other data-quality issues. The experiment will use Microsoft Power BI to connect to different data sources, load the data using Power Query, and perform data profiling using features such as Column Quality, Column Distribution, and Column Profile.

Dataset / Data Sources

The following data sources are considered for the experiment:

1. **Sales Data – CSV file**
 - Contains information related to sales transactions.
 - Includes fields such as Customer ID, Product, Quantity, Price, Date, and Sales Amount.
2. **Employee Database – Excel file**
 - Contains employee-related information.

- Includes fields such as Employee ID, Name, Department, Joining Date, Salary, etc.

3. Transaction Data – Database

- Contains transaction records stored in a database.
- Includes transaction-related fields such as Transaction ID, Customer ID, Transaction Date, Amount, and Transaction Type.

Objectives

The main objectives of this experiment are:

- To understand the process of data acquisition in BI tools.
- To connect Power BI with different data sources.
- To import and inspect datasets using Power Query.
- To understand the concept of data profiling.
- To identify missing, duplicate, inconsistent, and invalid data.
- To analyze the quality and distribution of data.
- To determine suitable data-cleaning and transformation requirements before analysis.

Tools Required

- Microsoft Power BI Desktop
- CSV dataset
- Excel dataset
- Database containing transaction data

Expected Output

After completing the experiment, the following outputs should be obtained:

- Successfully acquired data from CSV, Excel, and database sources.
- Data loaded into Power BI through Power Query.
- Data profiling results for the selected datasets.

- Identification of missing, invalid, inconsistent, or unusual values.
- A list of data-quality issues and their possible solutions.
- Clean and appropriately structured data ready for further analysis.